

Q.1: FIFO Page Replacement

```
#include <iostream>
using namespace std;

int main() {
    int n, frameSize;

    cout << "Enter number of pages in reference string: ";
    cin >> n;

    int ref[50];
    cout << "Enter the reference string: ";
    for (int i = 0; i < n; i++)
        cin >> ref[i];

    cout << "Enter number of frames: ";
    cin >> frameSize;

    int frames[10];
    int frameCount = 0; // how many frames are currently filled
    int front = 0;      // points to the oldest page (for FIFO replacement)
    int pageFaults = 0;

    // Initialize frames as empty (-1 means empty)
    for (int i = 0; i < frameSize; i++)
        frames[i] = -1;

    cout << "\nStep\tPage\tFrames\t\tStatus\n";

    for (int i = 0; i < n; i++) {
        int page = ref[i];
        bool hit = false;

        // Check if page is already in frames
        for (int j = 0; j < frameCount; j++) {
            if (frames[j] == page) {
                hit = true;
                break;
            }
        }

        if (!hit) {
            if (frameCount < frameSize) {
                // Free space available
                frames[frameCount] = page;
                frameCount++;
            } else {
                // Replace the oldest page (FIFO)
                frames[front] = page;
                front = (front + 1) % frameSize;
            }
            pageFaults++;
        }
    }
```

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    }

    // Print current state
    cout << i + 1 << "\t" << page << "\t";
    for (int j = 0; j < frameCount; j++)
        cout << frames[j] << " ";
    cout << "\t\t" << (hit ? "Hit" : "Fault") << endl;
}

cout << "\nTotal number of page faults = " << pageFaults << endl;

return 0;
}

```

Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/OS_lab_sem4/" && g++ lab_1.cpp -o lab_1 && "/Users/pratikthapamagar/OS_lab_se
m4/"lab_1
Enter number of pages in reference string: 10
Enter the reference string: 7 0 1 2 0 3 0 4 2 3
Enter number of frames: 3

Step    Page    Frames    Status
1       7       7         Fault
2       0       7 0       Fault
3       1       7 0 1     Fault
4       2       2 0 1     Fault
5       0       2 0 1     Hit
6       3       2 3 1     Fault
7       0       2 3 0     Fault
8       4       4 3 0     Fault
9       2       4 2 0     Fault
10      3       4 2 3     Fault

Total number of page faults = 9
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %

```

Q.2: Optimal (OPR) Page Replacement

```
#include <iostream>
using namespace std;

int main() {
    int n, frameSize;

    cout << "Enter number of pages in reference string: ";
    cin >> n;

    int ref[50];
    cout << "Enter the reference string: ";
    for (int i = 0; i < n; i++)
        cin >> ref[i];

    cout << "Enter number of frames: ";
    cin >> frameSize;

    int frames[10];
    int frameCount = 0;
    int pageFaults = 0;
    for (int i = 0; i < frameSize; i++)
        frames[i] = -1;

    cout << "\nStep\tPage\tFrames\t\tStatus\n";

    for (int i = 0; i < n; i++) {
        int page = ref[i];
        bool hit = false;

        // Check if page is already in frames
        for (int j = 0; j < frameCount; j++) {
            if (frames[j] == page) {
                hit = true;
                break;
            }
        }
        if (!hit) {
            if (frameCount < frameSize) {
                // Free space available
                frames[frameCount] = page;
                frameCount++;
            } else {
                // Find the page that will not be used for the longest time
                int farthest = -1, replaceIndex = -1;

                for (int j = 0; j < frameCount; j++) {
                    int k;
                    for (k = i + 1; k < n; k++) {
                        if (frames[j] == ref[k])
                            break;
                    }
                }
            }
        }
    }
}
```

```

        // If this page is never used again, replace it immediately
        if (k == n) {
            replaceIndex = j;
            break;
        }

        // Otherwise track the one used farthest in the future
        if (k > farthest) {
            farthest = k;
            replaceIndex = j;
        }
    }
    frames[replaceIndex] = page;
}
pageFaults++;
}

// Print current state
cout << i + 1 << "\t" << page << "\t";
for (int j = 0; j < frameCount; j++)
    cout << frames[j] << " ";
cout << "\t\t" << (hit ? "Hit" : "Fault") << endl;
}

cout << "\nTotal number of page faults = " << pageFaults << endl;

return 0;
}

```

Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/OS_lab_sem4/" && g++ lab_2.cpp -o lab_2 && "/Users/pratikthapamagar/OS_lab_sem4/"lab_2
Enter number of pages in reference string: 10
Enter the reference string: 7 0 1 2 0 3 0 4 2 3
Enter number of frames: 3

Step   Page   Frames   Status
1      7      7        Fault
2      0      7 0      Fault
3      1      7 0 1    Fault
4      2      2 0 1    Fault
5      0      2 0 1    Hit
6      3      2 0 3    Fault
7      0      2 0 3    Hit
8      4      2 4 3    Fault
9      2      2 4 3    Hit
10     3      2 4 3    Hit

Total number of page faults = 6
○ pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %

```

Q.3: LRU Page Replacement

```
#include <iostream>
using namespace std;

int main() {
    int n, frameSize;

    cout << "Enter number of pages in reference string: ";
    cin >> n;

    int ref[50];
    cout << "Enter the reference string: ";
    for (int i = 0; i < n; i++)
        cin >> ref[i];

    cout << "Enter number of frames: ";
    cin >> frameSize;

    int frames[10];
    int lastUsed[10]; // stores the step number each frame's page was last used
    int frameCount = 0;
    int pageFaults = 0;

    for (int i = 0; i < frameSize; i++) {
        frames[i] = -1;
        lastUsed[i] = -1;
    }

    cout << "\nStep\tPage\tFrames\t\tStatus\n";

    for (int i = 0; i < n; i++) {
        int page = ref[i];
        bool hit = false;
        int hitIndex = -1;

        // Check if page is already in frames
        for (int j = 0; j < frameCount; j++) {
            if (frames[j] == page) {
                hit = true;
                hitIndex = j;
                break;
            }
        }

        if (hit) {
            // Update its last used time
            lastUsed[hitIndex] = i;
        } else {
            if (frameCount < frameSize) {
                // Free space available
                frames[frameCount] = page;
                lastUsed[frameCount] = i;
            }
        }
    }
}
```

```

        frameCount++;
    } else {
        // Find the least recently used page (smallest lastUsed value)
        int minIndex = 0;
        for (int j = 1; j < frameCount; j++) {
            if (lastUsed[j] < lastUsed[minIndex])
                minIndex = j;
        }
        frames[minIndex] = page;
        lastUsed[minIndex] = i;
    }
    pageFaults++;
}

// Print current state
cout << i + 1 << "\t" << page << "\t";
for (int j = 0; j < frameCount; j++)
    cout << frames[j] << " ";
cout << "\t\t" << (hit ? "Hit" : "Fault") << endl;
}

cout << "\nTotal number of page faults = " << pageFaults << endl;

return 0;
}

```

Output:

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS

```

pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 % cd "/Users/pratikthapamagar/OS_lab_sem4/" && g++ lab_3.cpp -o lab_3 && "/Users/pratikthapamagar/OS_lab_sem4/"lab_3
Enter number of pages in reference string: 10
Enter the reference string: 7 0 1 2 0 3 0 4 2 3
Enter number of frames: 3

Step    Page    Frames    Status
1       7       7         Fault
2       0       7 0       Fault
3       1       7 0 1     Fault
4       2       2 0 1     Fault
5       0       2 0 1     Hit
6       3       2 0 3     Fault
7       0       2 0 3     Hit
8       4       4 0 3     Fault
9       2       4 0 2     Fault
10      3       4 3 2     Fault

Total number of page faults = 8
pratikthapamagar@Pratiks-MacBook-Pro OS_lab_sem4 %

```